

PERSONAL FINANCE & CREDIT RISK DATASET

Loan Approval Data Analysis

*A plain-language look at what separates an
approved loan application from a rejected one*

614

Loan Applications

422

Approved

68.7%

Overall Approval Rate

10x

Good vs Poor Credit Approval

Loan Approval Data Analysis

A plain-language look at what separates an approved loan application from a rejected one

About this report

This report looks at 614 real loan applications from a bank, using a public dataset that is widely used to practice credit-risk analysis (it is commonly known as the "Loan Prediction" dataset, and versions of it are published on Kaggle). Each row is one application, and it tells us about the applicant — their income, whether they are married, how many people depend on them, whether they are a graduate, whether they are self-employed, their credit history, and where they live — together with the loan they asked for and whether the bank approved it.

The goal is simple: read the data the way a bank's risk team would, clean it up properly, and work out, in plain terms, which factors actually move the needle on approval and which ones barely matter.

What's in the dataset

Column	What it means
Loan_ID	A unique reference number for the application
Gender, Married, Dependents	Basic household details of the applicant
Education	Graduate or not graduate
Self_Employed	Whether the applicant works for themselves
ApplicantIncome, CoapplicantIncome	Monthly income of the applicant and any co-applicant
LoanAmount	The loan amount requested, in thousands
Loan_Amount_Term	How many months the loan is to be repaid over
Credit_History	Whether the applicant has a clean repayment record (1) or not (0)
Property_Area	Rural, Semiurban or Urban
Loan_Status	The outcome we are trying to understand: approved or rejected

A look at the raw data

The first rows of the cleaned dataset, exactly as they sit in the workbook before any grouping or charting.

Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
LP001002	Male	No	0	Graduate	No	5849	0		360		1 Urban	Y
LP001003	Male	Yes	1	Graduate	No	4583	1508	128	360		1 Rural	N
LP001005	Male	Yes	0	Graduate	Yes	3000	0	66	360		1 Urban	Y
LP001006	Male	Yes	0	Not Graduate	No	2583	2358	120	360		1 Urban	Y
LP001008	Male	No	0	Graduate	No	6000	0	141	360		1 Urban	Y
LP001011	Male	Yes	2	Graduate	Yes	5417	4196	267	360		1 Urban	Y
LP001013	Male	Yes	0	Not Graduate	No	2333	1516	95	360		1 Urban	Y
LP001014	Male	Yes	3+	Graduate	No	3036	2504	158	360		0 Semiurban	N
LP001018	Male	Yes	2	Graduate	No	4006	1526	168	360		1 Urban	Y
LP001020	Male	Yes	1	Graduate	No	12841	10968	349	360		1 Semiurban	N
LP001024	Male	Yes	2	Graduate	No	3200	700	70	360		1 Urban	Y
LP001027	Male	Yes	2	Graduate		2500	1940	109	360		1 Urban	Y
LP001028	Male	Yes	2	Graduate	No	3073	8106	200	360		1 Urban	Y
LP001029	Male	No	0	Graduate	No	1853	2840	114	360		1 Rural	N
LP001030	Male	Yes	2	Graduate	No	1299	1086	17	120		1 Urban	Y
LP001032	Male	No	0	Graduate	No	4950	0	125	360		1 Urban	Y
LP001034	Male	No	1	Not Graduate	No	3596	0	100	240		Urban	Y
LP001036	Female	No	0	Graduate	No	3510	0	76	360		0 Urban	N
LP001038	Male	Yes	0	Not Graduate	No	4887	0	133	360		1 Rural	N
LP001041	Male	Yes	0	Graduate		2600	3500	115			1 Urban	Y
LP001043	Male	Yes	0	Not Graduate	No	7660	0	104	360		0 Urban	N
LP001046	Male	Yes	1	Graduate	No	5955	5625	315	360		1 Urban	Y
LP001047	Male	Yes	0	Not Graduate	No	2600	1911	116	360		0 Semiurban	N
LP001050		Yes	2	Not Graduate	No	3365	1917	112	360		0 Rural	N
LP001052	Male	Yes	1	Graduate		3717	2925	151	360		Semiurban	N
LP001066	Male	Yes	0	Graduate	Yes	9560	0	191	360		1 Semiurban	Y

Figure 0. Sample rows of the raw loan application data, as originally recorded.

Objective

Before touching the data, it helps to be precise about what we are trying to find out. This analysis was built around five questions:

- How many of the 614 applications were approved, and what does the overall approval rate look like?
- Does an applicant's credit history actually change their chances, and by how much?
- Does income — the applicant's own income, or the combined household income — make approval more likely?
- Does the size of the loan requested, or where the property is located, affect the outcome?
- Do personal factors such as education, marital status or self-employment matter, or are they mostly noise?

The intended audience is anyone who wants a quick, honest read of what drives loan approval in this dataset — a student building a portfolio piece, or someone learning what a bank's screening process tends to reward. The output is three things: a cleaned dataset, a set of Excel charts built entirely from live formulas, and this written report.

Approach

The analysis follows the usual order: understand the raw data, clean it, then analyse it group by group. Every chart in this report was generated natively inside Excel (not pasted in as a picture from another tool) and every number behind it is a formula that reads from the cleaned data, so the workbook recalculates if the underlying data changes.

Scope of this report

This is a descriptive analysis, not a predictive model. It explains patterns in the 614 applications we have — it does not build a scoring formula to approve or reject new applicants. Where a group is small (for example, applications for very large loans), we say so, because a small group's approval rate can swing a lot from just one or two decisions.

Data cleaning process

Real application forms are rarely filled in perfectly, and this dataset is no exception — 7 of the 13 columns had at least one missing value, totalling 149 blank cells. Rather than deleting incomplete rows (which would have thrown away good information from every other column in that row), each column was fixed on its own terms. Every fix below is visible in the workbook's Cleaning_Log sheet, and is done with a formula, not a one-off manual edit.

Column	Missing	How it was handled
Credit_History	50	Kept as its own "Not Available" group, instead of guessing. Credit history turned out to matter so much for the outcome that guessing it could have hidden the real pattern.
LoanAmount	22	Filled with the median loan amount (128 thousand). The median resists being pulled up by a few very large loan requests, unlike an average.
Self_Employed	32	Filled with the most common answer ("No"), since the large majority of applicants are not self-employed.
Dependents	15	Filled with the most common answer ("0" dependents).
Loan_Amount_Term	14	Filled with the most common term (360 months, i.e. 30 years).
Gender	13	Filled with "Not Stated" rather than guessed, since there was no reliable way to infer it.
Married	3	Filled with the most common answer ("Yes"). Only 3 rows were affected.

Other checks

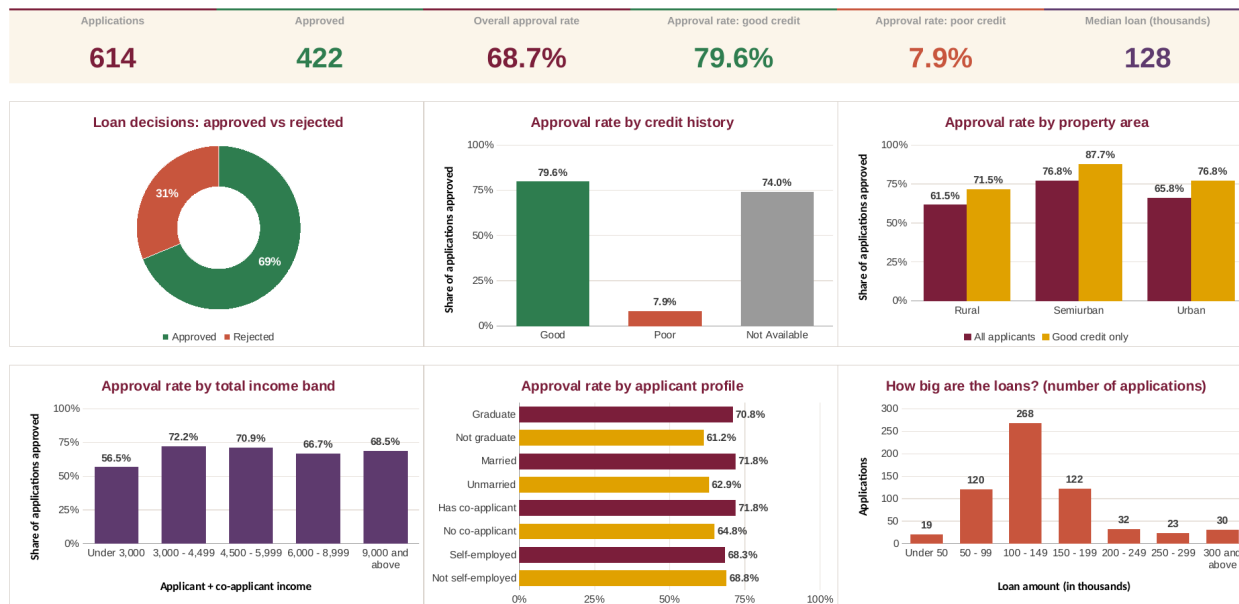
- **Duplicate applications:** none found — every Loan_ID appears exactly once.
- **Row count:** no rows were removed. All 614 applications are still in the cleaned data.
- **Consistency:** Loan_Status was recoded from Y/N to Approved/Rejected, and Credit_History from 1/0 to Good/Poor, so that every chart reads clearly without a lookup key.
- **New columns added:** Total_Income (applicant income plus co-applicant income) and grouped income/loan bands, so we could compare like-sized groups fairly.
- **Unusually large values:** a standard statistical check (values more than 1.5 times the interquartile range above the typical range) flagged 50 incomes, 18 co-applicant incomes and 41 loan amounts as unusually high. These look like genuine high earners and large loans rather than data-entry errors, so they were kept in the data, and medians were used instead of averages wherever these values could otherwise distort the picture.

The dashboard at a glance

Everything below is one Excel dashboard, built with native Excel charts driven by formulas. The six charts cover the outcome itself, the two factors that matter most (credit history and property area), and four further angles (income, loan size, and applicant profile) checked to see whether they add anything on top.

Loan Approval Dashboard

614 bank loan applications: what decides who gets approved?



Source: Loan Prediction dataset (614 applications). All figures come from formulas on the Analysis sheet.

Figure 1. Full loan approval dashboard, built in Excel from the cleaned dataset.

Finding 1: credit history is the single biggest factor

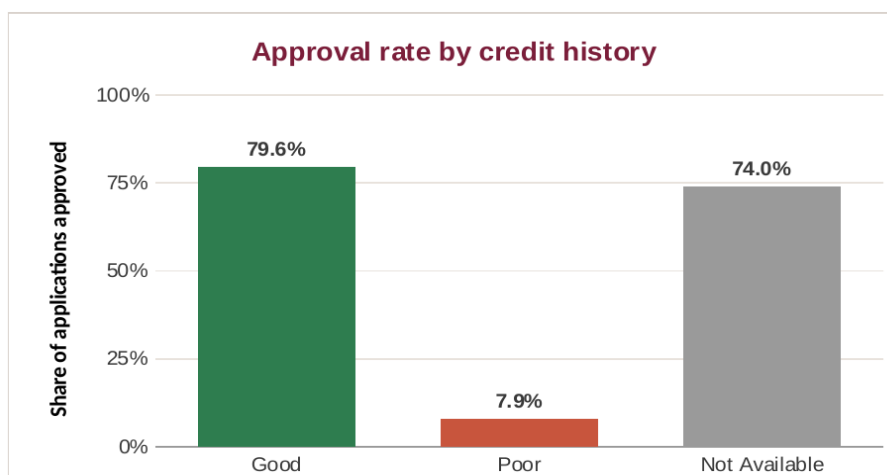


Figure 2. Approval rate by credit history.

What the data shows

Applicants with a good credit history are approved **79.6%** of the time; applicants with a poor credit history are approved just **7.9%** of the time — a ten-times difference, and by far the strongest pattern in the dataset. The 50 applicants with no credit history on file were still approved 74.0% of the time, suggesting the bank leaned on other information for them.

Finding 2: location shifts the odds, but far less than credit history

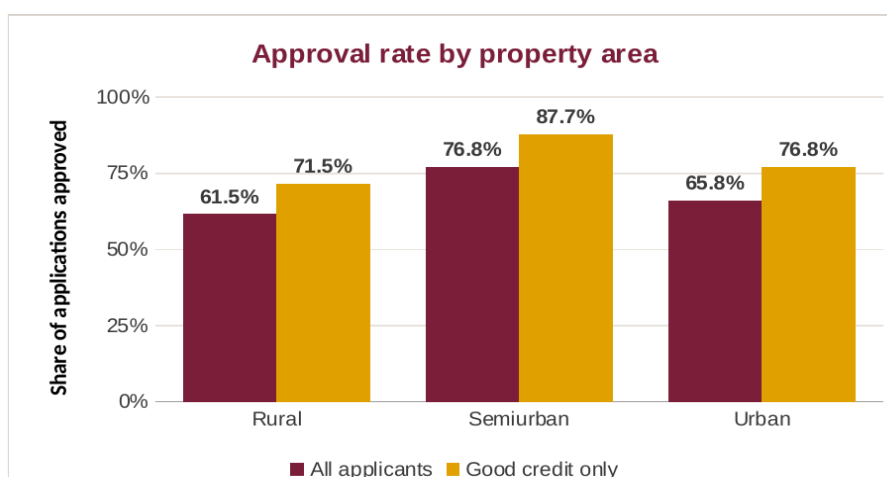


Figure 3. Approval rate by property area, all applicants vs. good-credit applicants only.

What the data shows

Semiurban properties have the highest approval rate (76.8%), ahead of urban (65.8%) and rural (61.5%) properties. This pattern holds even among applicants who already have good credit (87.7% semiurban vs. 71.5% rural), so location is doing real work here, not just standing in for credit history.

Three more angles, checked and mostly ruled out

Income, personal profile and loan size were checked next, to see whether any of them add much on top of credit history and location. Short answer: a little, but not much.

Finding 3: income barely changes the outcome

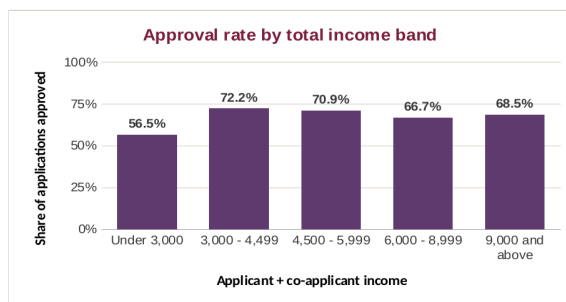


Figure 4. Approval by income band.

Approval rates sit in a narrow 56.5%-72.2% range across every income band, with no clear upward trend as income rises. A higher income alone does not reliably buy a higher approval rate here.

Finding 4: personal factors, a small effect

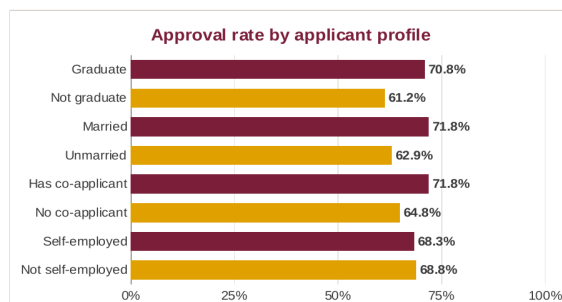


Figure 5. Approval by applicant profile.

Graduates, married applicants and those with a co-applicant are approved 7-9 points more often than their counterparts. Self-employment makes almost no difference (68.3% vs. 68.8%).

Finding 5: the loan amount itself is not a strong predictor

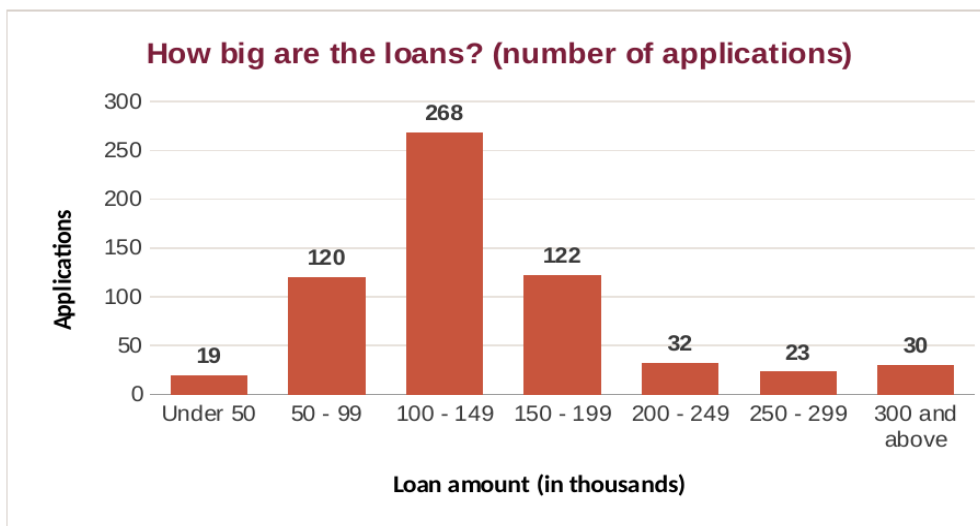


Figure 6. Number of applications by loan amount, in thousands.

Most applications (268 of 614) ask for between 100,000 and 149,000, and approval rates drift only gently downward as the loan gets larger — from around 79% for the smallest loans to 60-66% for the largest, though the largest bands have very few applications and should be read as indicative rather than precise.

Conclusion

Out of 614 applications, 422 were approved — an overall approval rate of 68.7%. Once the data is broken down, one factor stands well above the rest.

614 Applications	422 Approved	68.7% Overall rate	79.6% Good credit	7.9% Poor credit
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What we found

- **Credit history decides almost everything.** A good credit history takes approval odds from around 8% to almost 80%. This is by far the strongest pattern in the data.
- **Property location plays a real, secondary role.** Semiurban applicants are approved noticeably more often than rural or urban applicants, even after holding credit history constant.
- **Income, education, marital status and self-employment matter, but only at the margins.** Each shifts approval odds by a handful of percentage points — useful context, not a deciding factor on its own.
- **The size of the loan requested has only a mild effect,** with larger loans approved slightly less often, though the largest bands have too few applications to be fully confident in the exact numbers.

What this means in practice

If you had to guess whether a single application would be approved and could only ask one question, “is the credit history good?” would tell you far more than any other question in this dataset. Everything else — income, education, location, loan size — adds nuance around that central fact, but does not come close to overriding it.

Limitations

This dataset has 614 rows, so groups that look interesting but are small (for example, the largest loan band, with 30 applications) should be treated as a general signal rather than an exact number. The data also does not tell us why an application was rejected beyond these variables, so factors such as debt levels or the bank's internal policy at the time are not visible here. Finally, this is a historical snapshot: it describes how this batch of applications was decided, not a guarantee of how future applications will be treated.